

Glomeruli detection

Outline

Accurate identification and classification of glomeruli are essential for diagnosing diabetic kidney disease and guiding clinical decision-making.

Aim

This project aims to develop a deep learning pipeline capable of:

1. Identifying glomeruli.
2. Segmenting the identified glomeruli.
3. Classifying the glomeruli into distinct categories using an **unsupervised** approach.

Given the lack of ground truth data for glomeruli classification, the project will focus on exploring the separability of glomeruli classes based on levels of necrotization (see below figure). The unsupervised method will help reveal potential patterns or clusters within the data.

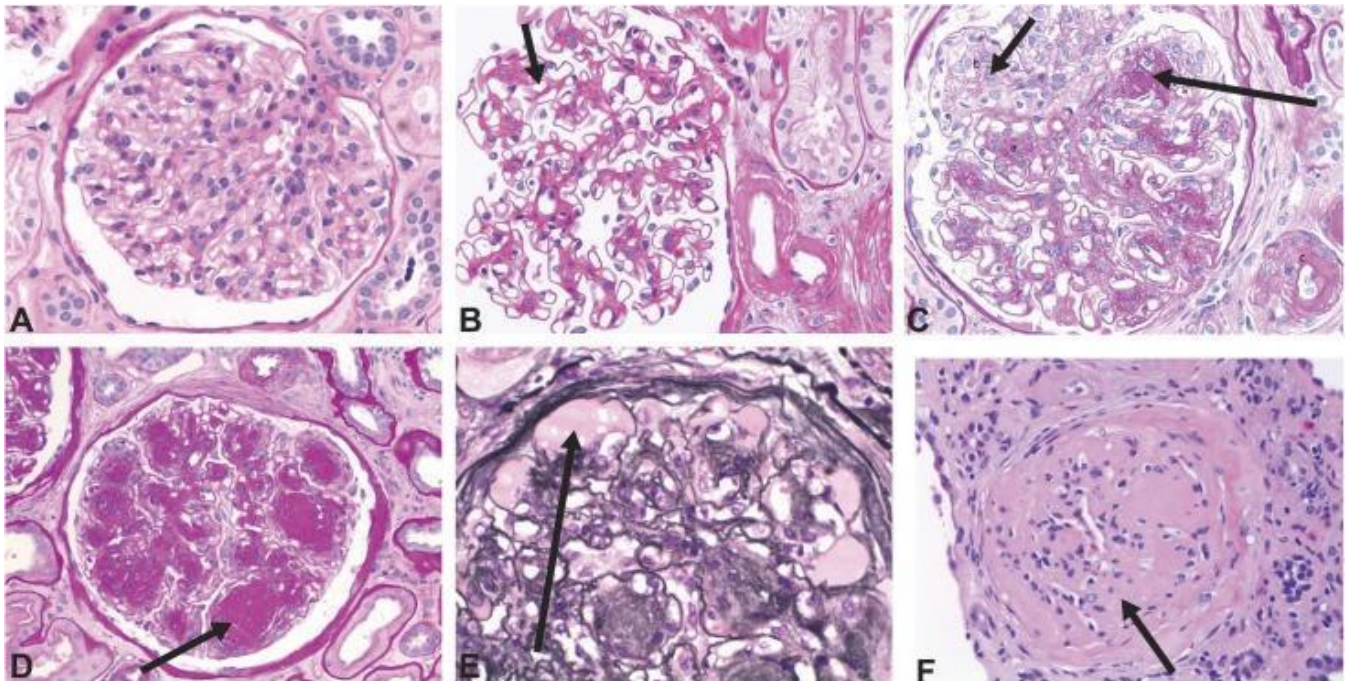


Figure 1. Increasing level of glomeruli's necrotization (from A, healthy glomerulus, to F, complete dysplasia).

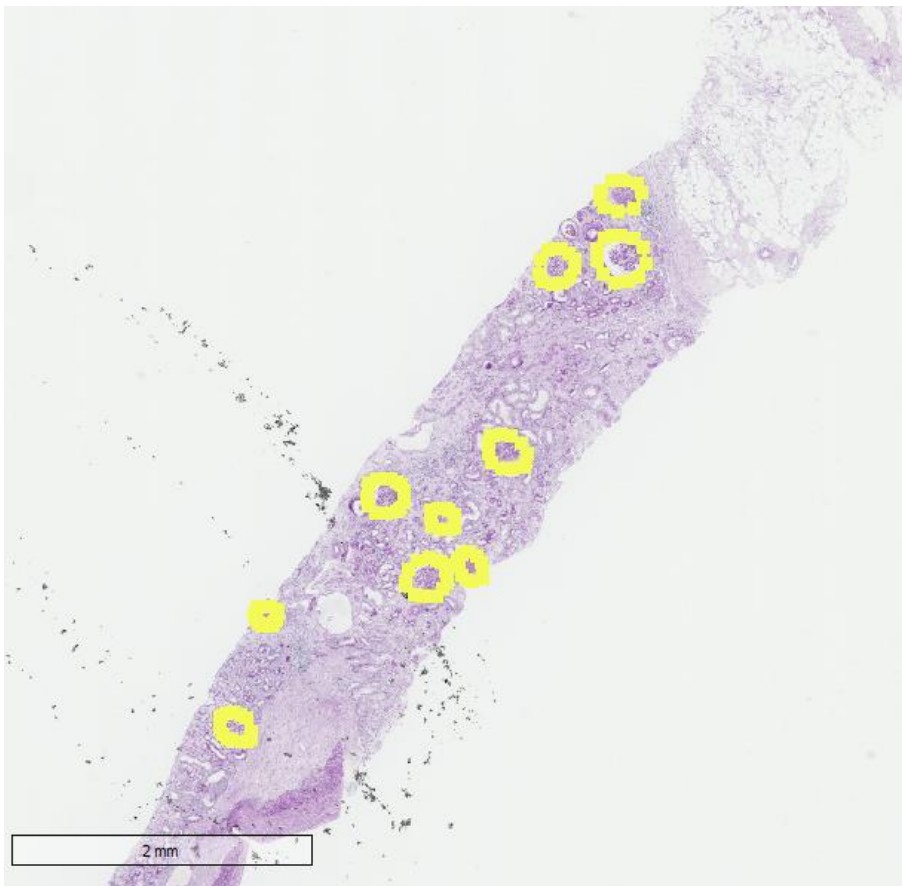
Dataset

Available [here](#).

Such dataset includes:

- 1) WSIs of biopsies from kidney tissue (.svs)
- 2) The corresponding annotation showing the glomeruli (.xml)

Below is an example of a WSI and the corresponding annotations.



Software

To visualise images and annotations consider the [ASAP](#) software.

Furthermore, several packages built upon [OpenSlide](#) exist to help WSIs management via python. For instance, <https://github.com/ysbecca/py-wsi>.